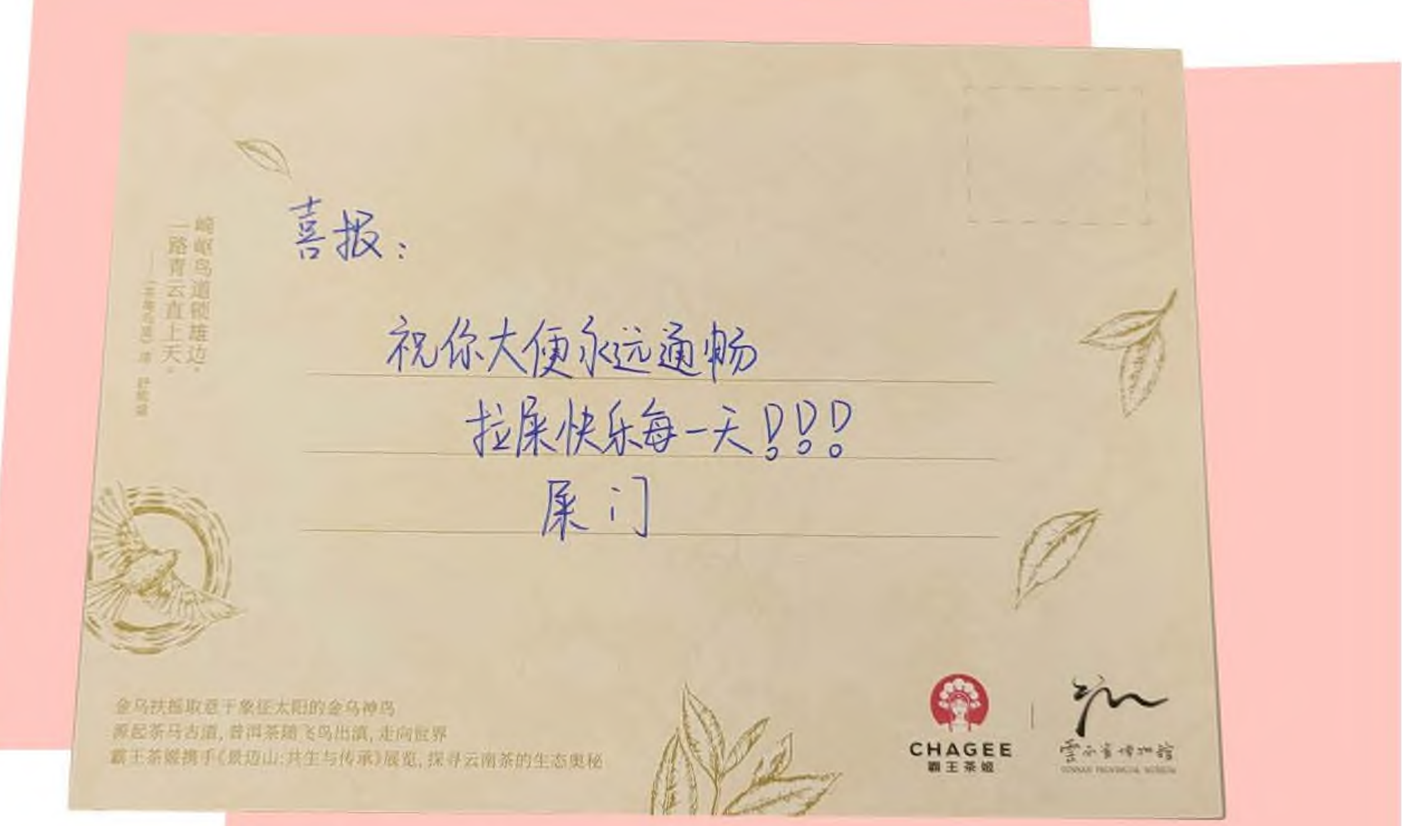
moodoo

“ Use the Bathroom,
Make a Friend. ”

- **Area:**
  Toilet Installation, HCI, Arduino
- **Feature:**
  Experiment, health
- **Duration:**
  Aug.2024 - Oct.2024
- **Solo project**

MOTIVATION



Good News:
Wishing you smooth and easy bowel movements!
May you enjoy your time in the restroom every day!

On my second-year birthday, I received many birthday cards. Some had typical birthday wishes, while others contained more interesting ones. One card that said, "Wishing you smooth bowel movements forever" left a strong impression on me. Everyone present loved this wish and found it very practical.

Later, I noticed more and more similar wishes about "smooth bowel movements" in other places. What seemed like a routine detail of going to the bathroom was, in fact, a significant part of life, and people's attention to it was beyond what I had imagined.

Having a pleasant bathroom experience is very important to me and many others. However, the current experience still seems to have a lot of room for improvement. So, I wondered—could I make many improvements and explore new ways for people to interact with toilets?

How might we design a better restroom experience that enhances user satisfaction?

Name: Wen Tian
Age: 20
Occupation: Undergraduate Student
Health Status: Chronic lower back pain from prolonged sitting, shoulder pain, tendinitis

In that case, the toilet seat won't be cold anymore

After all, the toilet seat comes into direct contact with the human body. Often, when I use a public restroom or someone else's toilet, it can be quite uncomfortable. If this airbag were exclusively mine, I wouldn't have to worry about the hygiene of the toilet seat

The shape feels a bit uncomfortable; it seems too far back. When I use the toilet, I can't reach the back part. The overall design could definitely be improved

The process of adjusting based on posture on-site, followed by retesting after adjustments

I usually spend a short time in the bathroom, so the feedback might need to be more immediate, but I'm not too concerned about sitting for long periods

I often experience diarrhea, which is quite uncomfortable. It would be great if it could make me feel a bit more at ease

Name: Jia Yuan
Age: 28
Occupation: Doctoral Student
Health Status: Mild constipation, chronic lower back pain from prolonged sitting, tendinitis

Name: Lian Cheng
Age: 55
Occupation: Corporate Employee
Health Status: Long-term hemorrhoids, occasional constipation

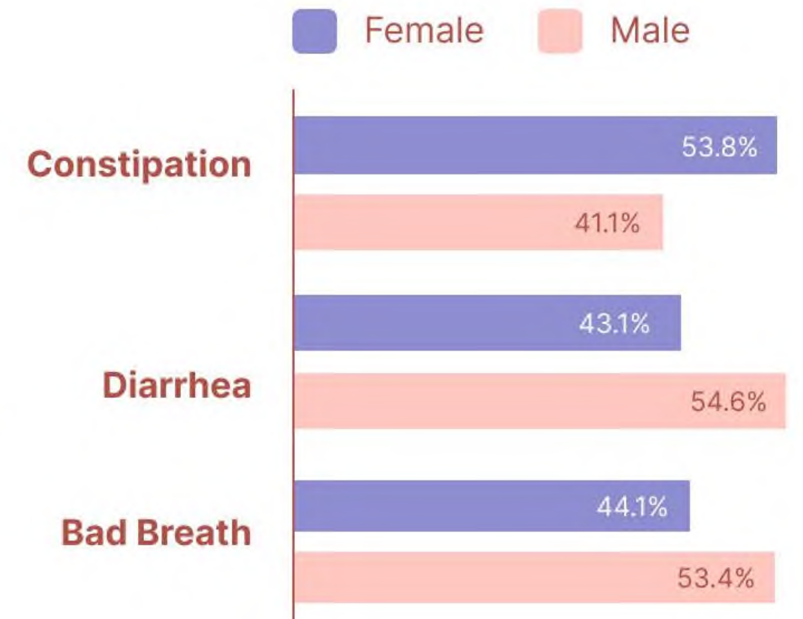
I generally take a long time in the bathroom, and of course, using my phone can extend that time even more. So, a reminder from the toilet could be quite interesting

I've also dealt with hemorrhoids for many years. Sometimes it's painful, so if the toilet could help me find a more comfortable position, that would be very convenient

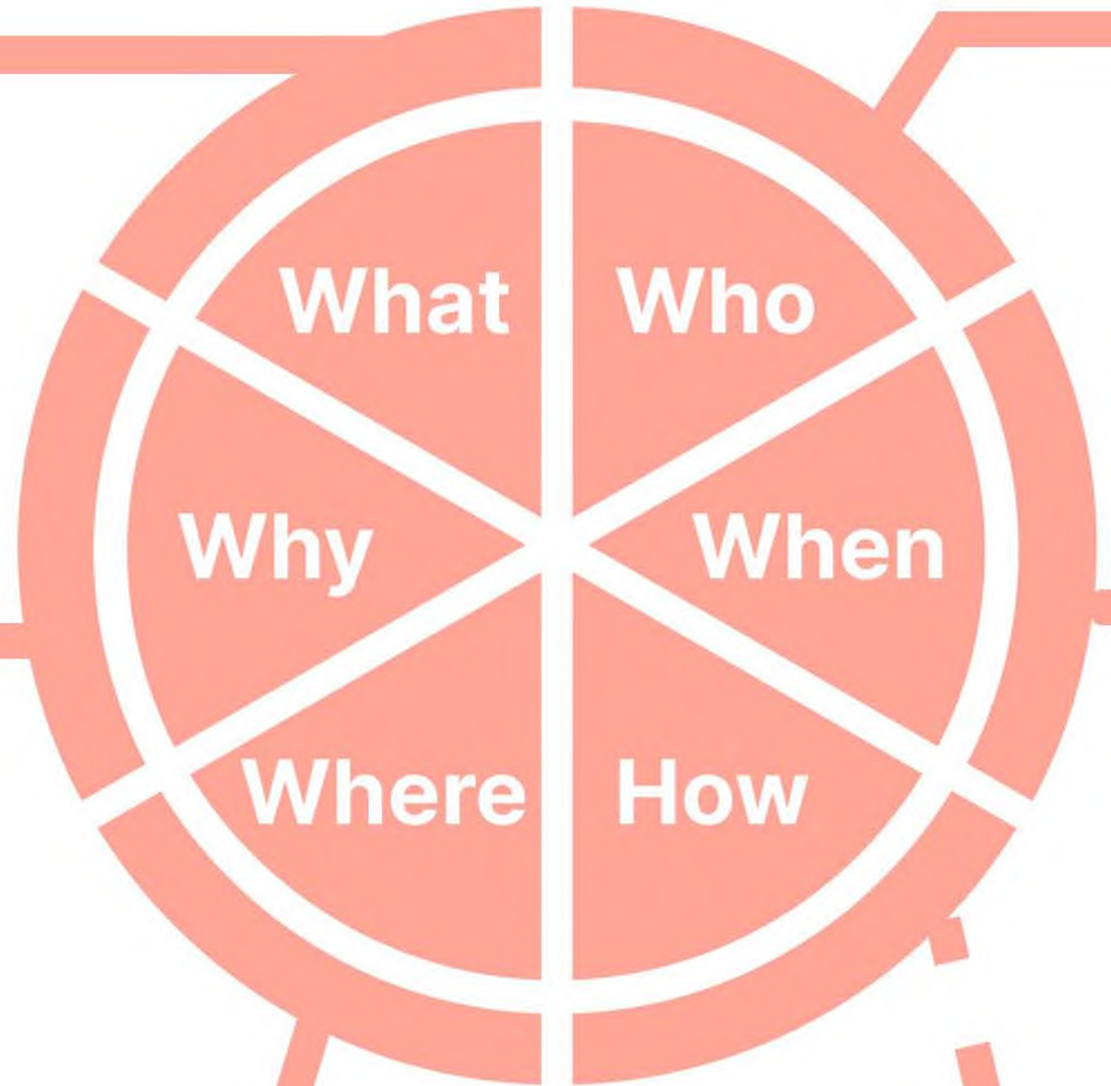
Currently, smart toilets mainly make washing the backside very easy and comfortable, but the seat itself isn't very pleasant to sit on

Every morning after I wake up, I spend about half an hour in the bathroom; it's almost like my entertainment time

According to the 2021 Gut Health White Paper, over 90% of people are currently suffering from or have experienced gastrointestinal discomfort, with insufficient attention given to it. Gut issues are primarily focused on diarrhea and constipation, and can also lead to bad breath.



Function Definition - 5W1H



What do I want to achieve?

We aim to enhance the toilet experience by personalizing attention to each user's physical health and psychological well-being. Our goal is to address existing issues associated with toilet use.

Why do I want to achieve a solution?

Since toilets have become the mainstream form of sanitation, problems such as constipation and prolonged sitting have been on the rise. Public restrooms also pose significant hygiene challenges.

Where does it impact?

The primary application context is the restroom, particularly those equipped with toilets. This includes private toilets in homes and public toilets. Squat toilets are not included.

Who is affected?

Target users include office workers, students, individuals with constipation, hemorrhoid sufferers, travelers, and those with various musculoskeletal issues or in a state of sub-health.

When did the issue arise?

As toilets have become the main form of sanitation, issues like constipation and prolonged sitting have increased, along with hygiene problems in public restrooms.

How can I involve relevant people?

The demographic affected by these issues is quite broad. We can explore various age groups with different health concerns through surveys, interviews, and field testing.

Issues & Objectives

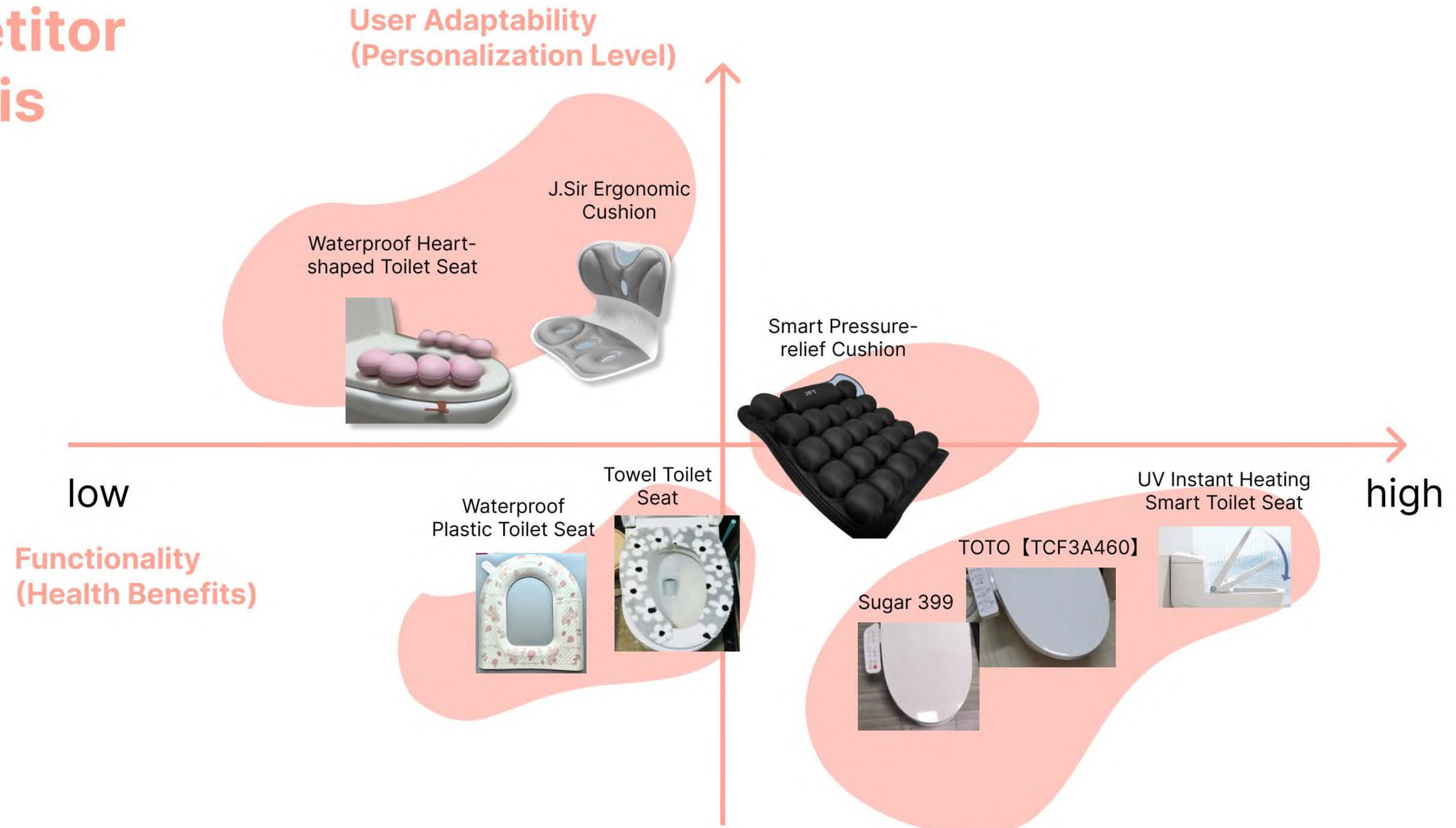
Through research and analysis, we aim to identify common issues that most people experience during toilet use, as well as what toilet time means to them.

Many people find using the toilet to be a dull experience, often focusing their attention on their phones.

Many individuals adopt incorrect postures while using the toilet and tend to sit for extended periods.

Using the toilet can be a relaxing time, providing a brief escape from work-related stress.

Competitor Analysis



Function \ Product	UV Instant Heating Smart Toilet Seat	TOTO TCF3A460 Smart Toilet Seat	J.Sir Ergonomic Cushion	Waterproof Plastic Toilet Seat
Personalized Smart Assistant	●●●○○	●●●●○	●○○○○	●○○○○
Constipation/ Period Care	●●●●○	●●●○○	●●○○○	●●○○○
Smart Control	●●●○○	●●●●○	●●○○○	●○○○○
User Comfort	●●●○○	●●●○○	●●●●○	●●●○○

Ideations

Option 1 Leg Strap + Air Pump

- Pressure Detection
- Air Pump Actuation
- Manual Adjustment

Concept Evaluation

- Could lead to increased discomfort or numbness in one leg
- Requires additional wear
- Position may be too close to the toilet rim, increasing risk of contamination

Option 2 Backrest Behind Toilet + Servo

- Pressure Detection
- Servo Actuation
- Direct Adjustment

Concept Evaluation

- Users often cannot make direct contact
- Limited adjustable directions reduce the potential for varied user experiences
- Inconvenient to assess skin through clothing, making it difficult to gauge user status directly

Option 3 Strap on Waist + Shape Memory Alloy

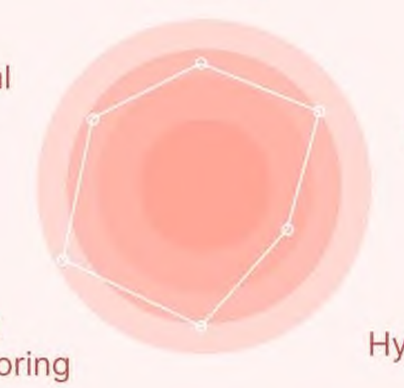
- Contact Detection
- Shape Memory Alloy
- Manual Adjustment

Concept Evaluation

- Excessive contact area encircles the entire waist, causing discomfort in a seated position
- Manual adjustments are inconvenient and not easily accessible
- Changing the position can put pressure on the waist

Option 4 Seat Cushion Air Pad + Air Pump

- Pressure Detection
- Air Pump Actuation
- Direct Adjustment



Concept Evaluation

- Allows direct contact with the skin for more accurate measurements, but does not require special wear
- Located underneath the body for easy manual adjustments
- Various postural changes can be achieved by altering directions

MoodBoard

Inflatable Texture

The surface texture is fluffy and soft, providing a close and warm feel.

Overall Seat Shape

The overall appearance resembles a seat, designed to be ergonomic, ensuring comfort and relaxation when sitting down.

Modular Assembly

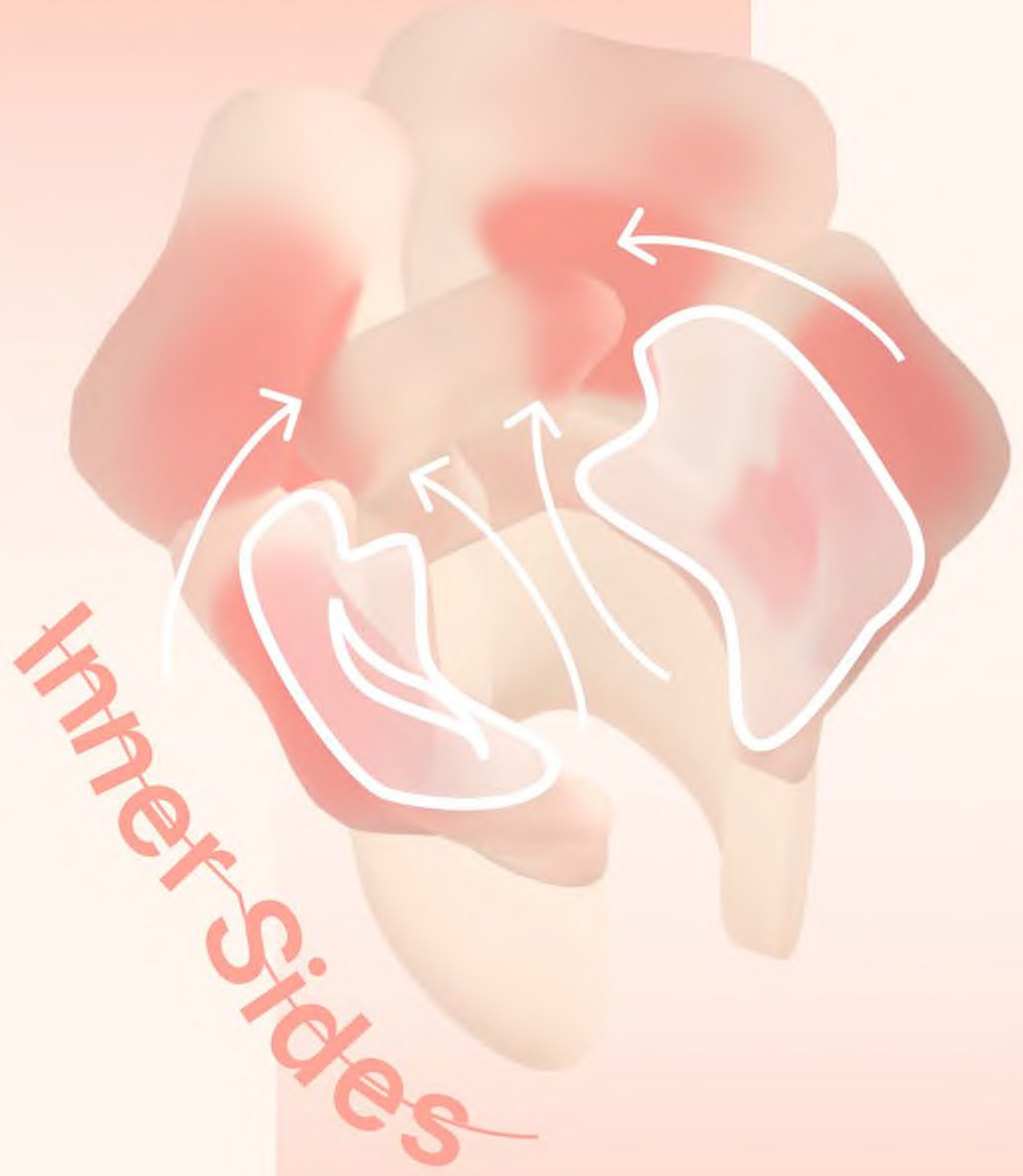
Multiple small modules combine to form a cohesive whole, maintaining individual independence while achieving a visually appealing unity. The small modules allow for greater flexibility in adjustments.
4o mini

Irregular Sensation

The specific details of the shape draw inspiration from the layering of multiple irregular volumes, resulting in a unique irregular form.

Concept

This section has two states of variation. In the first state upon activation, the front expands outward and upward, allowing the user to lean back comfortably.



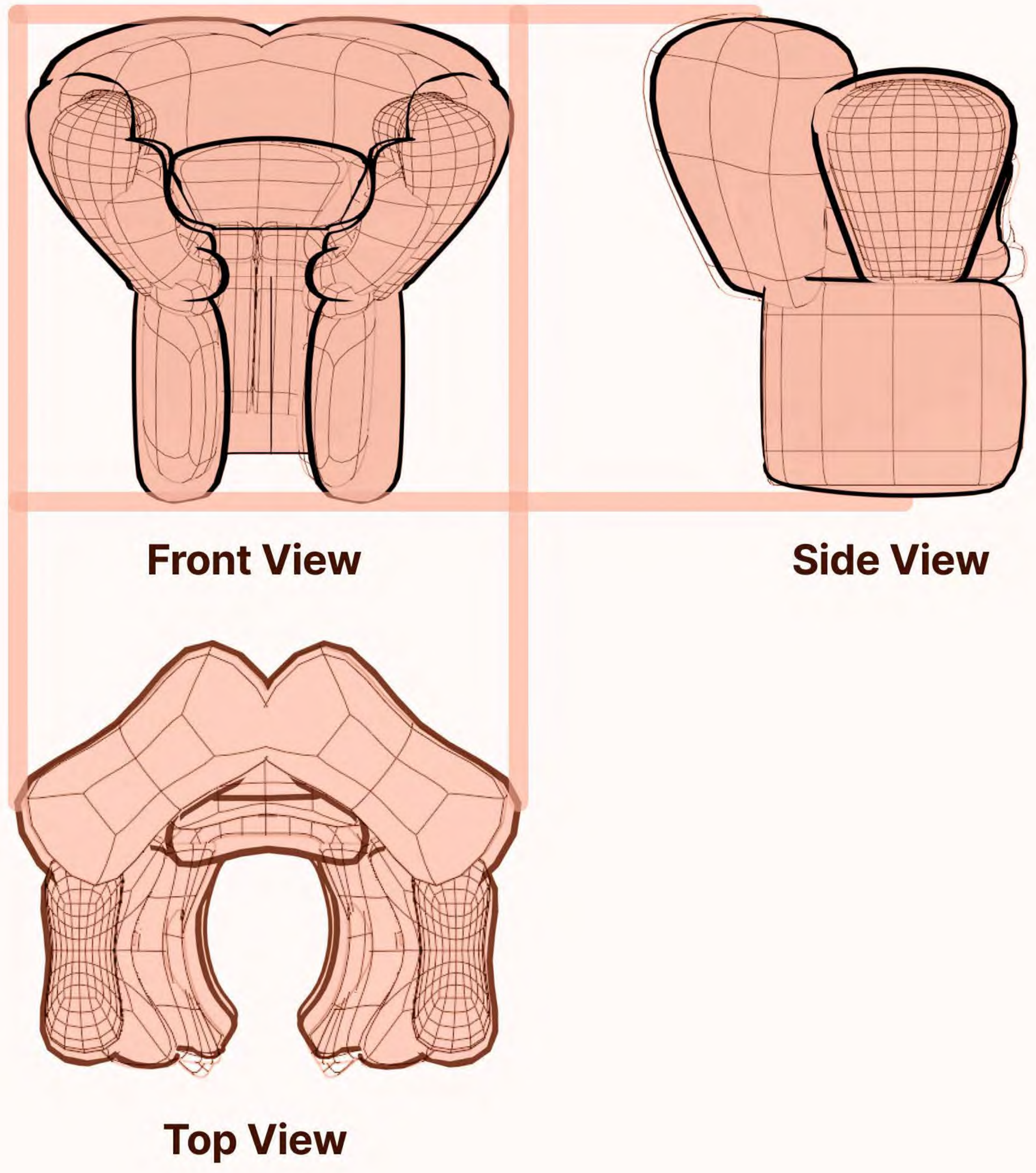
In the second state upon activation, the rear expands inward and upward, providing support for the user's back, making it easier to lean forward.

When inactive, users do not make direct contact with this area. Upon activation, the backrest elevates forward, offering better support for the user's lower back.

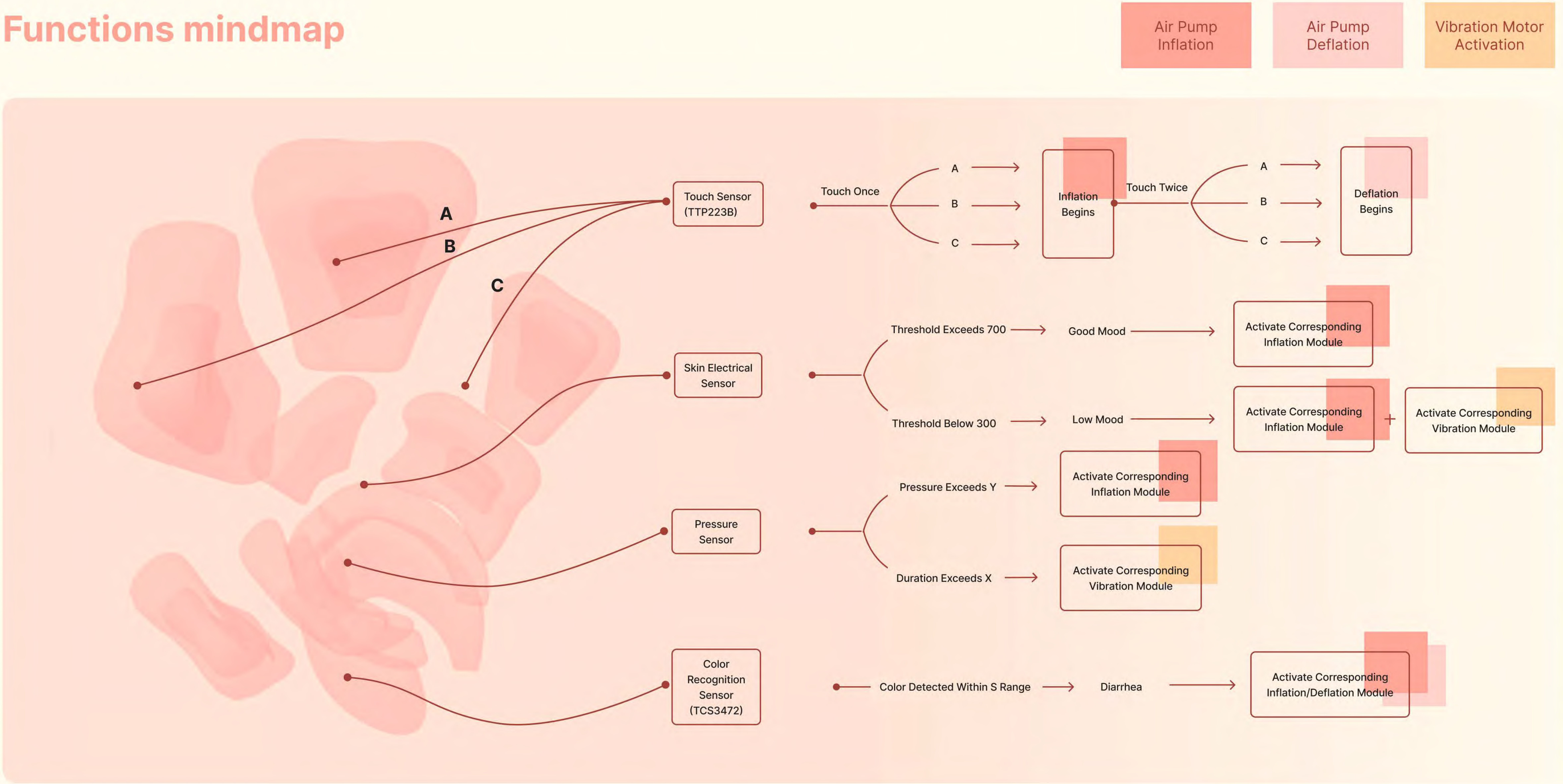
When activated, the sides expand outward while the front pushes forward, providing ample tactile feedback and support from the rear, making it easier to change positions and enhance comfort.

When activated, the upper airbag expands inward and backward, enveloping the user's waist and facilitating adjustments for various postures.

Orthographic Views

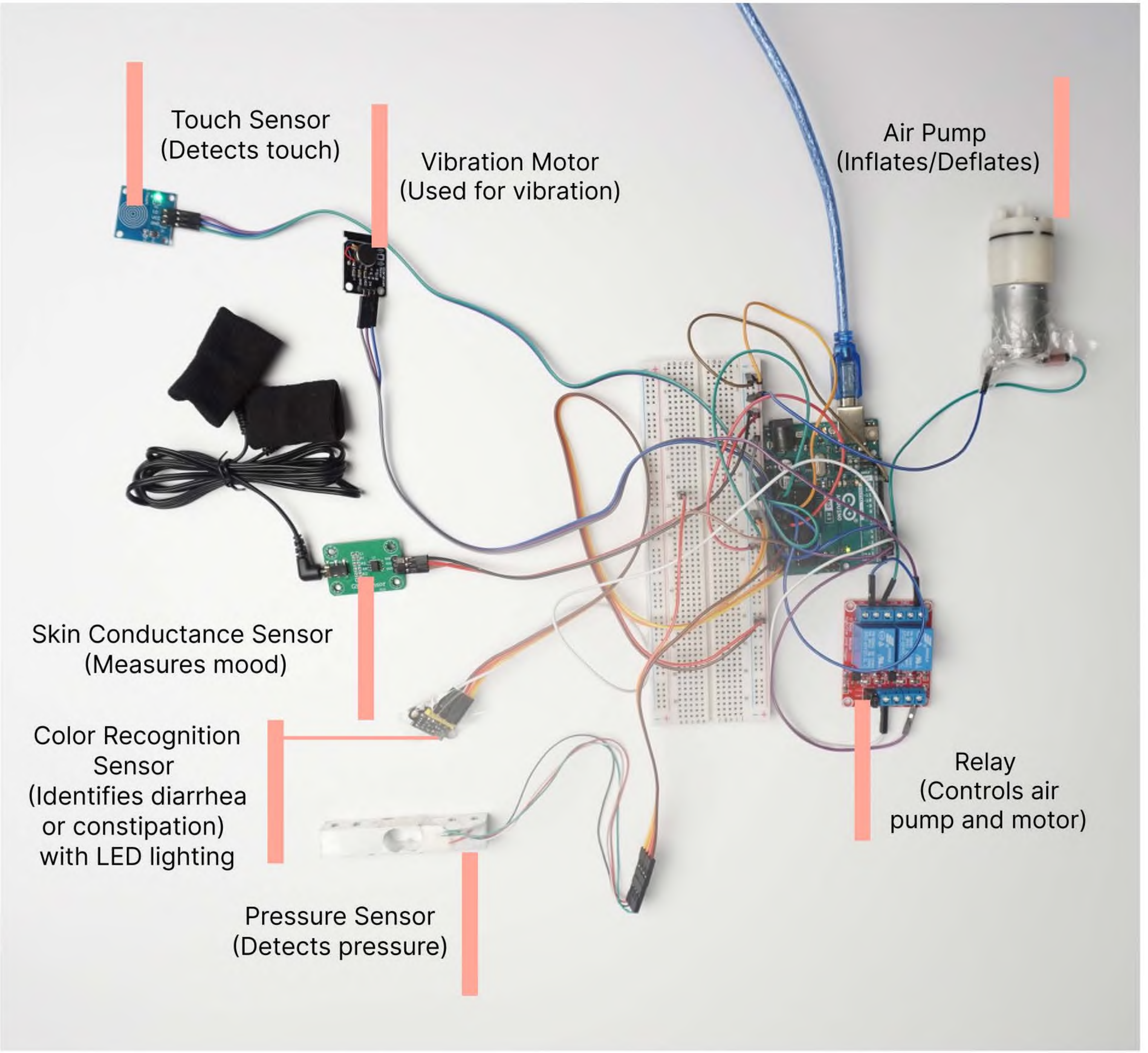


Functions mindmap



Functional prototype

Based on the product definition above, I implemented it using Arduino code and wiring. By measuring in different directions, corresponding changes can be achieved. Although it is not possible to fully replicate the inflation and deflation effect, the challenges lie in sensor measurement errors and the inability of the air pumps and sensor modules to support changes across all parts. However, it is expected that in the future, by increasing the number of air pumps and sensors, a combination can be established to achieve better results.



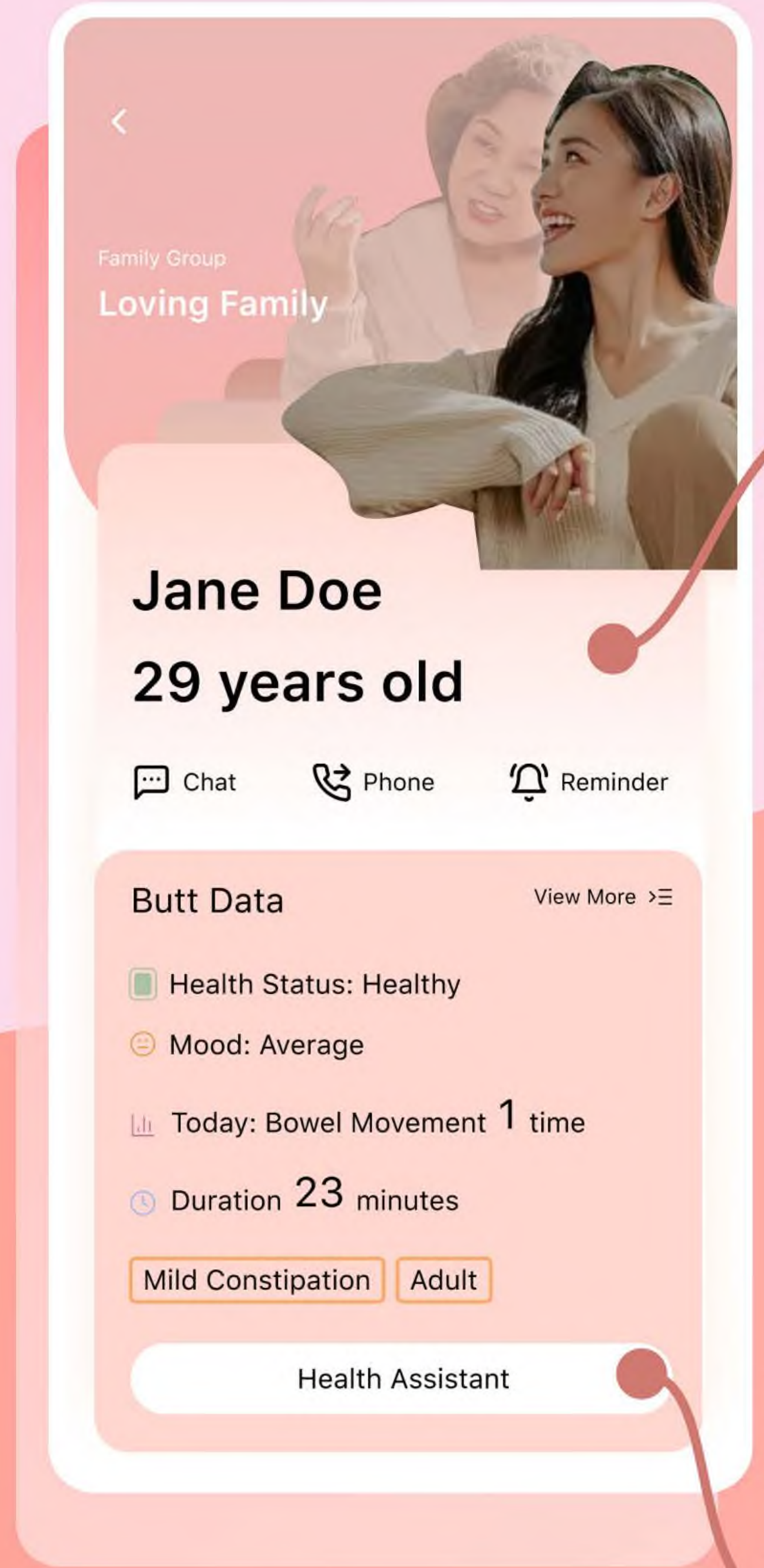

```
#include <HX711.h>

HX711 scale;  // 创建 HX711 对象

const int touchPin = 2;         // 触摸传感器引脚
const int pumpPin = 4;          // 气泵控制引脚
const int vibrationPin = 3;      // 震动马达控制引脚
const int colorSensorPin = A2;   // 颜色传感器引脚
const int gsrPin = A1;           // GSR 传感器引脚

float weight;
unsigned long pressureStartTime = 0;
bool isPumping = false;
bool isVibrating = false;

void setup() {
  Serial.begin(9600);
  scale.begin(A0, A1);  // 连接 HX711
  pinMode(touchPin, INPUT);
  pinMode(pumpPin, OUTPUT);
  pinMode(vibrationPin, OUTPUT);
  digitalWrite(pumpPin, LOW);
  digitalWrite(vibrationPin, LOW);
}

void loop() {
  weight = scale.get_units(10);  // 获取重量

  // 功能 1: 根据压力自动充放气
  if (weight > 30.0) {
    startPump();
  } else {
    stopPump();
  }

  // 功能 2: 触摸控制
  if (digitalRead(touchPin) == HIGH) {
    if (!isPumping) {
      startPump();
      isPumping = true;
    } else {
      stopPump();
      isPumping = false;
    }
    delay(1000);  // 防止多次触摸
  }

  // 功能 3: 检测心情低落（通过 GSR 传感器判断）
  int gsrValue = analogRead(gsrPin);
  if (gsrValue < 500) {  // 自定义心情低落的阈值
    startVibration();
  } else {
    stopVibration();
  }

  // 功能 4: 持续压力检测
  if (weight > 30.0) {
    if (pressureStartTime == 0) {
      pressureStartTime = millis();
    }
    if (millis() - pressureStartTime > 600000) {  // 10分钟
      startVibration();
    }
  } else {
    pressureStartTime = 0; // 重置计时
    stopVibration();
  }

  // 功能 5: 检测腹泻（假设通过颜色识别传感器）
  int colorValue = analogRead(colorSensorPin); // 假设读取的值
  if (colorValue < 100) {  // 腹泻颜色的阈值
    startPump();
  }

  Serial.println(weight);  // 输出当前重量
}

void startPump() {
  digitalWrite(pumpPin, HIGH);
}

void stopPump() {
  digitalWrite(pumpPin, LOW);
}

void startVibration() {
  digitalWrite(vibrationPin, HIGH);
}

void stopVibration() {
  digitalWrite(vibrationPin, LOW);
}
```

Work-like prototype

Once inflated, the shape will appear as follows. During the manual production process, I used flesh-colored silk fabric as the filling inside the airbag, inflatable silicone airbags as the main body, and two layers of frosted TPU fabric for the outer layer. The production method involves wrapping a foam board of appropriate size with the silk fabric, placing it inside the airbag, and then inflating the airbag. Different-sized airbags are attached together using tape according to different modules. Finally, two layers of TPU fabric are wrapped around the airbag and adhered with tape.

FUTURE SCENARIOS

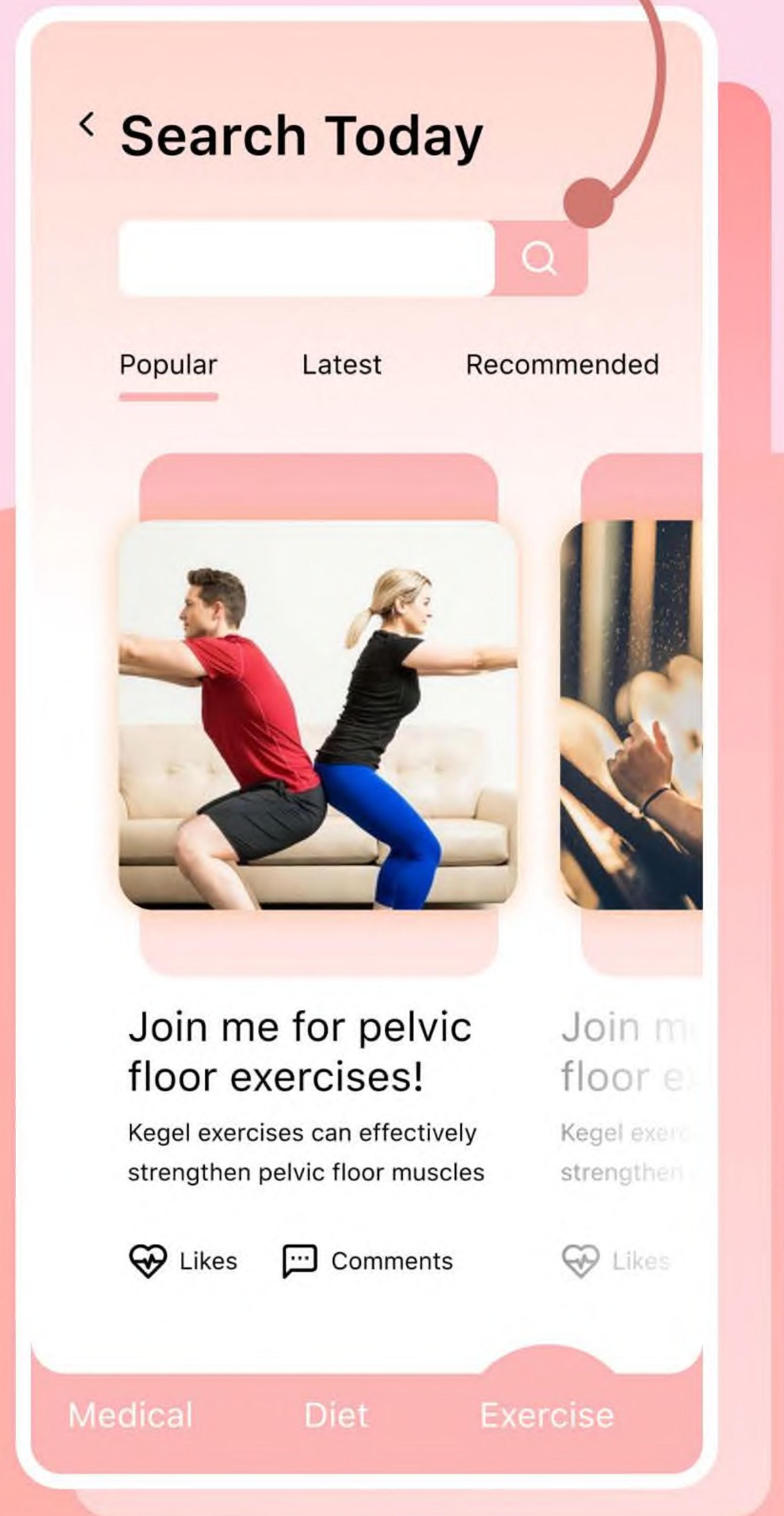
For the future development of this design, I have made bolder predictions and explored various aspects, including its appearance and functionality

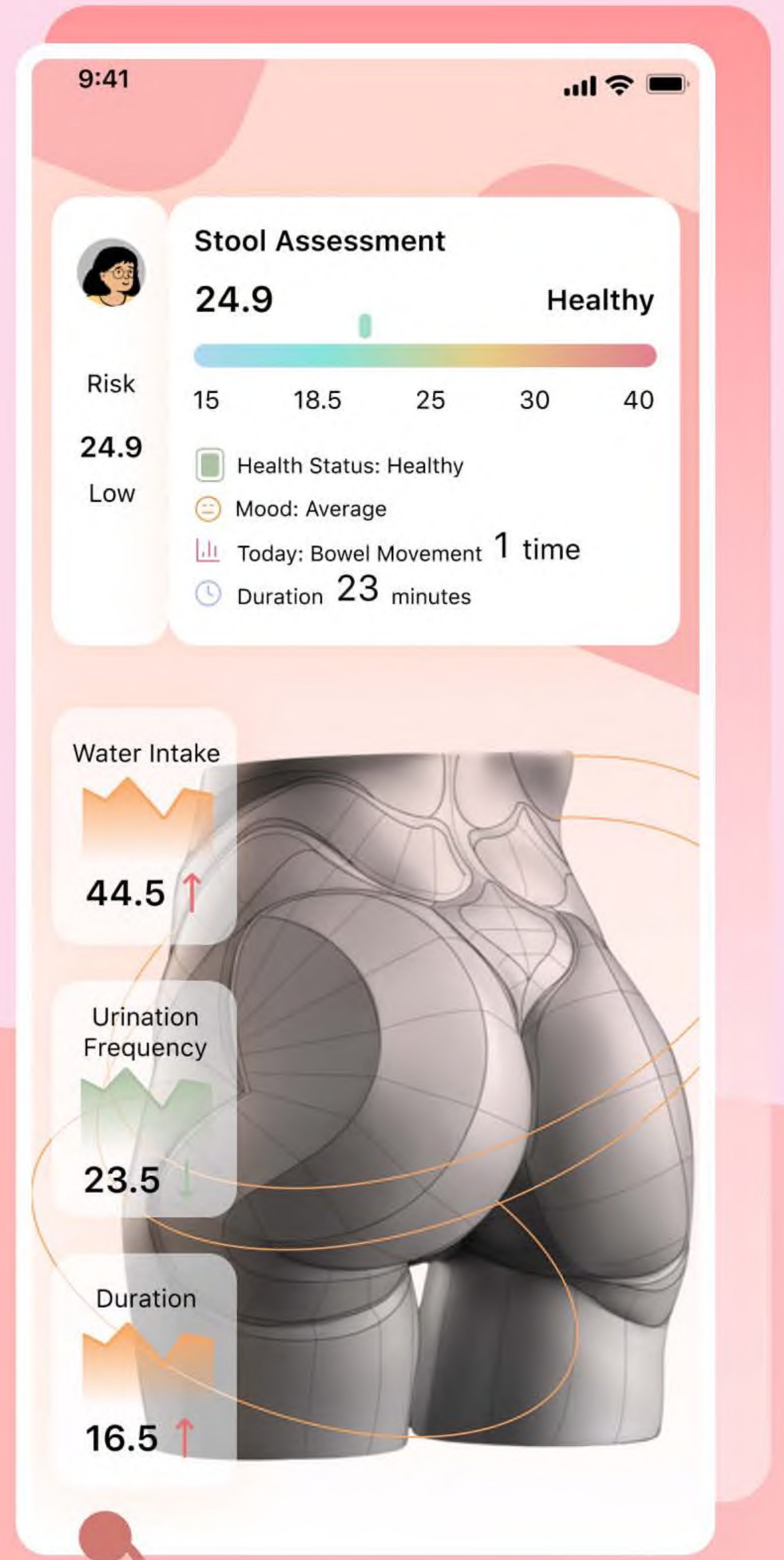


Real-time monitoring of family members' gastrointestinal health

View detailed data and communicate with family members at any time

Access relevant health information, or post inquiries and view content shared by others





View detailed statistics and analysis, with visualizations of daily bodily changes and timely reminders

External Fixing Strap

The soft material can conform to different body types, providing a better fit and secure attachment

Side Airbags

The airbags inflate on both sides to form a customized toilet seat during use, enhancing hygiene and safety for personal use

Bottom Material

The bottom material continuously monitors and alerts for bodily changes, tracking various health conditions and abnormal data

Leg Fibers

The fibers in the legs help better support posture adjustment and assist with positioning during toilet use